

		needed.
24	B	The force per unit length acting on Z by X, $\frac{F_{XZ}}{l_Z} = B_X l_Z = \frac{\mu_0 l^2}{2\pi (2d)} = F$ The force per unit length acting on Z by Y, $\frac{F_{YZ}}{l_Z} = B_Y l_Y = \frac{3\mu_0 l^2}{2\pi d} = 6F$ Currents in Y and Z are in the same direction: Y attracts Z. Currents in X and Z are in opposite direction: X repels Z. Hence, the net force of $5F$ towards Y.
25	C	The copper disc “sees” the magnet rotating. The portion of the disc either “sees” an approaching pole or a